Baseline results

Run 20261024_210638 | code: snapshot commit 6f9c01e; the run recorded 1f231d2 plus uncommitted changes | AMD Ryzen 7 7735HS with Radeon Graphics | one run, simulation only | built by evaluation/baseline_pdf.py from the run's files

1. Verdict at a glance

The robot localizes well and maps the interior fully, but it cannot finish the mission, walks slowly, and does not run in real time on this machine. Every number below was measured in this run; a class (Good, Improve, Poor) is assigned from the thresholds in section 2.

Component	Good	Improve	Poor
Exploration (Frontier, Planning, Locomotion agents)	1	1	4
Localization (SLAM agent)	3	2	2
Mapping (Mapping agent)	3	3	3
Agents and real-time operation	3	3	3
All metrics	**10**	**9**	**12**

Where we are	Evidence from this run
Good	No runaway drift: final error 0.24 % of 158 m, never above 63 cm; heading error 0.74 deg; wall recall 91 %; SLAM update 10.6 ms; scan matching, loop closure and the heading prior each clearly help (section 2b).
Can improve	Absolute error 31 cm (target 15 cm); wall accuracy 12.9 cm; frontier selection time; memory use.
Poor	Mission not completed; speed 0.088 m/s; 46 % of the path walked after coverage plateaued; rooms re-entered 2.2 times each; relative error per metre 7.4 %; real-time factor 0.32; control-loop stalls up to 2.2 s.

2. Our results: what each value is, why it came out this way, and its class

Each table lists one component. **Value** is measured. **Class rule** shows the limits: for example '<= 15 good, > 50 poor' means 15 or less is Good, above 50 is Poor, in between is Improve. **Basis** says where the limit comes from: *published* (anchored on a cited number), *budget* (a real-time budget of this system) or *target* (an engineering target proposed here, not a literature value).

Run: timeout (1800s) before all frontiers were cleared; simulated 1800 s in 5666 s of wall time; 158.0 m walked; 6 of 8 rooms completed. Ground truth is the simulator pose; the map reference is a grid built with that pose, not an independent floor plan.

Exploration (Frontier, Planning, Locomotion agents)

Metric	Value	Class	Class rule	Basis	What it is	Why this value
Task completed	no	Poor	-	task	The mission reported that every reachable frontier was explored.	It ended at the 1800 s cap with 6 of 8 rooms completed. The interior was already mapped around t = 930 s; the 48 frontiers left are mostly not real (outside or at the outer wall) and cannot be cleared, so completion is never reported.
Average speed	0.088 m/s	Poor	>= 0.5 good, < 0.2 poor	published	Ground-truth distance divided by simulated time.	The calibrated gait peaks near 0.079 m/s in the simulator, and the robot also turns, stops and revisits. Reference: ANYmal 0.75 m/s normal walking, Spot 1.6 m/s max (product pages); FARE's indoor runs average 1.78 m/s (distance/ time, derived).
Distance walked after coverage plateaued	46 % of path	Poor	<= 10 good, > 25 poor	target	Share of the path walked after the explored area reached 97 % of its final value (t = 930 s).	73.2 m were walked after the plateau for only 2.1 percentage points more coverage: the robot kept chasing frontiers that cannot be cleared (fixes F1, F2).
Room entries per room	2.25	Poor	<= 1.25 good, > 2 poor	target	Room-entered events divided by rooms found. About 1 means each room is entered once.	18 entries for 8 rooms: the room labels change as the map grows, so rooms are 'entered' again and re-swept (fix F3).
Stuck events	2.5 per 100 m	Improve	<= 1 good, > 3 poor	target	Times the anti-stuck watchdog fired.	4 events in 158 m; each cost a reverse step and a dropped target.

Unreachable targets / paths planned	0.12 % of planner calls	Good	<= 2 good, > 5 poor	target	Targets the A* planner could not reach.	9 unreachable targets; they come from phantom frontiers beyond walls.

Localization (SLAM agent)

Metric	Value	Class	Class rule	Basis	What it is	Why this value
ATE as % of path length	0.20 %	Good	<= 0.3 good, > 1 poor	published	ATE RMSE divided by distance walked. Lets runs of different length be compared.	Published ANYmal D missions (145-332 m): ORB-SLAM3 stereo 0.14 %, FAST-LIVO2 0.04 % (derived from mean mission length). Our path is long because the robot revisits a small arena, which flatters this ratio.
ATE RMSE	31.0 cm	Improve	<= 15 good, > 50 poor	published	Typical position error over the run, against simulator ground truth, no alignment.	Odometry error accumulates and 68 loop closures pull it back; the largest error was 63 cm. It exceeds the 15 cm map tolerance. Published long-mission ATE: 11-33 cm (ANYmal D, outdoors).
Final drift	0.24 % of path	Good	<= 0.5 good, > 2 poor	target	Position error at the end of the run divided by the distance walked.	38.5 cm after 30 minutes: no runaway drift.
Largest position error	63 cm	Improve	<= 50 good, > 100 poor	target	Worst single-step error.	Never above 1 m, so the estimate stayed usable for navigation throughout.
Mean heading error	0.74 deg	Good	<= 2 good, > 5 poor	target	Orientation error against ground truth.	Gyro-led yaw with a wall-direction (Manhattan) prior keeps heading small; removing the prior raises the error to 18.8 deg (section 2b).
Relative error per metre (RPE)	7.4 % of 1 m	Poor	<= 2 good, > 5 poor	target	Error of the motion over each 1 m of true path, independent of accumulated drift.	Odometry plus gyro alone scores 1.8 %, so scan matching adds local jitter while fixing global drift (fix F7).
Scan-match acceptance (diagnostic)	26.9 %	Poor	>= 80 good, < 50 poor	target	Share of SLAM updates whose scan match passed its gates; the rest fall back to odometry.	The gates against odometry are tight (0.01 m plus the step length). Rejection reasons are not recorded, so this is an indicator, not a failure rate (fix F16).

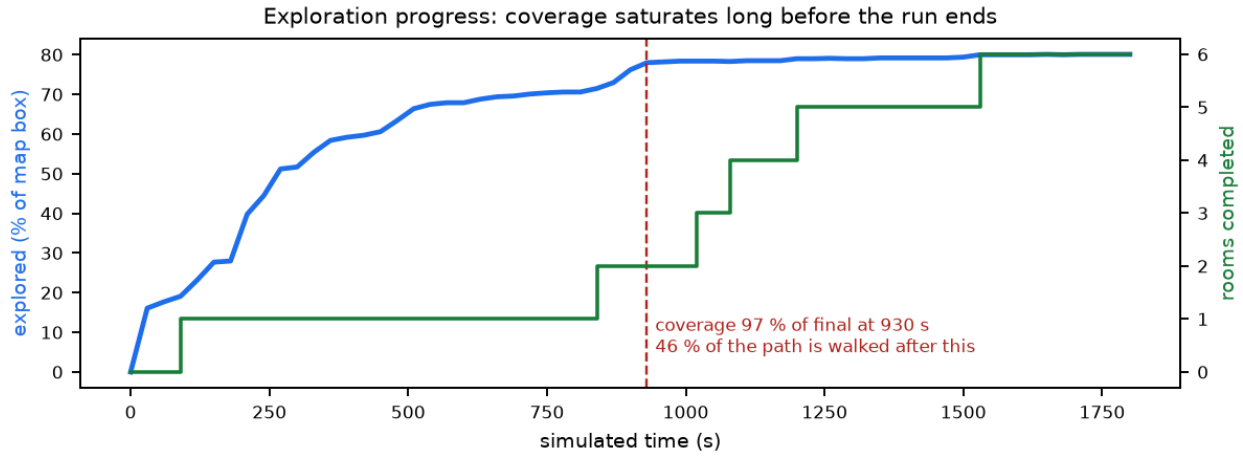
Mapping (Mapping agent)

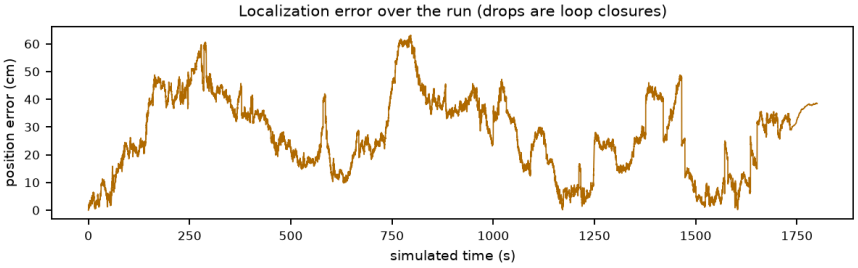
Metric	Value	Class	Class rule	Basis	What it is	Why this value
Wall accuracy	12.9 cm	Poor	<= 5 good, > 10 poor	published	Mean distance from each SLAM wall cell to the nearest reference wall cell.	Walls are thick: 5407 occupied cells against 3184 in the reference built from the same rays with the true pose. Pose error (ATE 31 cm) smears walls over several 5 cm cells. Published neural 3D maps reach 1.6-2.9 cm, a different representation.
Wall completion	6.4 cm	Improve	<= 5 good, > 10 poor	published	Mean distance from each reference wall cell to the nearest SLAM wall cell.	Walls are present; the error is thickness, not missing walls.
Completion ratio (5 cm)	54.1 %	Poor	>= 90 good, < 70 poor	published	Share of reference wall cells matched within 5 cm (one cell).	Low because walls are offset by more than one cell. Published neural SLAM on Replica: 90-97 % (different scenes and definition).
Precision	66.8 %	Poor	>= 85 good, < 70 poor	target	Share of SLAM occupied cells that match the reference within 15 cm.	Drops when walls are doubled or smeared and when free-space rays leak past walls.
Recall	91.3 %	Good	>= 85 good, < 70 poor	target	Share of reference occupied cells found by SLAM within 15 cm.	The robot finds almost every wall it saw.
F1	77.2 %	Improve	>= 85 good, < 70 poor	target	Balance of precision and recall.	Held back by precision.
Free-space IoU	81.2 %	Improve	>= 85 good, < 70 poor	target	Overlap of free cells with the reference.	Free space is mostly right; leaks beyond walls reduce it.

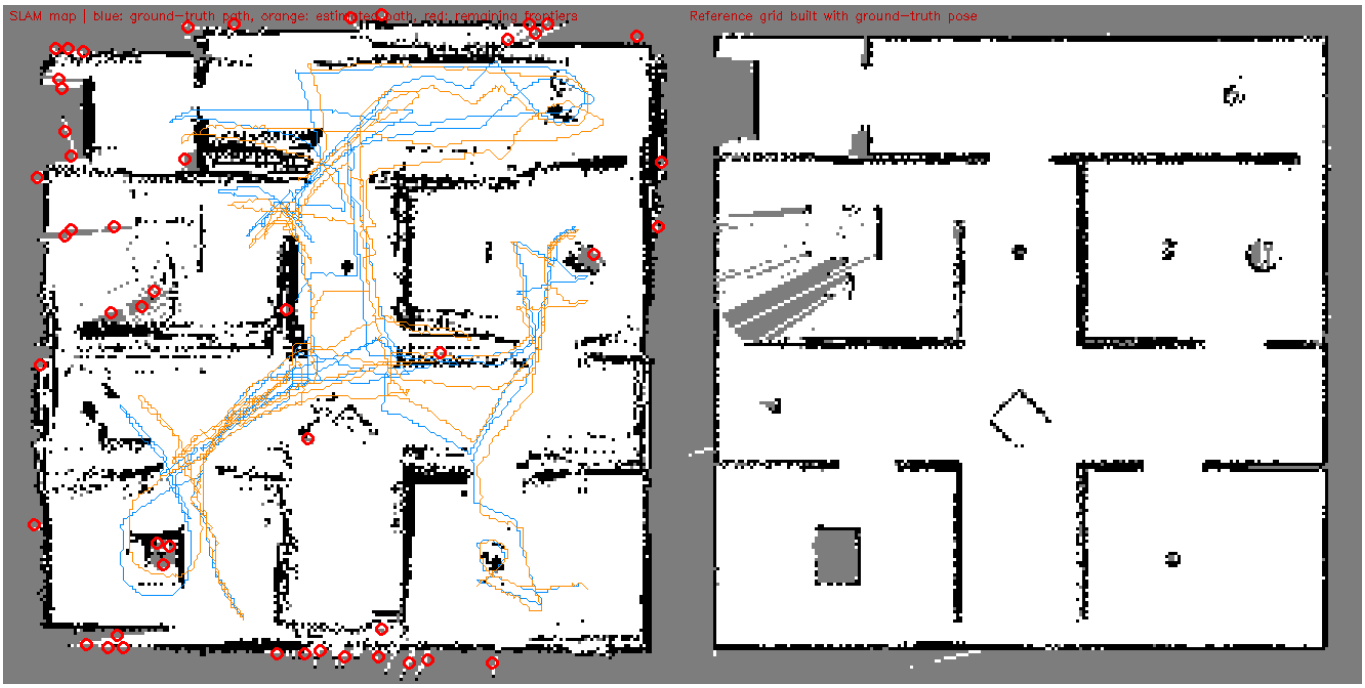
Map update time (p95)	36.9 ms	Good	<= 100 good, > 200 poor	budget	Time to integrate one depth frame (10 Hz budget 100 ms).	From the agent's own timing of each frame update on the 220 x 220 grid.
Map memory	0.19 MB	Good	<= 25 good, > 100 poor	published	Occupancy grid size.	A 2D grid is tiny; published neural scene models use 20-40 MB (MNE-SLAM Table 5; not like-for-like).

Agents and real-time operation

Metric	Value	Class	Class rule	Basis	What it is	Why this value
SLAM update time (median)	10.6 ms	Good	<= 20 good, > 50 poor	published	Time per SLAM update (scan match and loop checks).	CPU only. Published tracking: ORB-SLAM2 RGB-D 25.58 ms/frame, MNE-SLAM 22.1 ms (different tasks and hardware).
Frontier selection (median)	155.0 ms	Improve	<= 100 good, > 300 poor	budget	Time per target selection (1 Hz).	A geodesic search in pure Python; the most expensive per call (fix F14).
Planner time (p95)	127.6 ms	Improve	<= 100 good, > 250 poor	budget	A* path planning at 5 Hz.	Median 0.4 ms but p95 128 ms: the calls that run a full A* search are slow.
Slowest single agent call	1986 ms	Poor	<= 200 good, > 1000 poor	budget	Longest stall of any agent.	Perception reached 1986 ms and Planning 1061 ms; causes not yet traced (fix F13).
Control-loop step (p95)	380 ms	Poor	<= 50 good, > 100 poor	budget	Wall time of one whole control step (budget 50 ms at 20 Hz).	The agents use 46 % of wall time; the rest is simulator stepping, image transfer, HUD and per-step map queries (split not measured).
Real-time factor	0.32 x	Poor	>= 1 good, < 0.5 poor	budget	Simulated seconds per wall second; 1 or more is real time.	Not real time on this laptop (fix F12).
Process CPU (mean)	86 % of one core	Good	<= 100 good, > 200 poor	budget	CPU use of the Python process (simulator excluded).	One thread of Python work. Published for reference: ORB-SLAM3 stereo 290 %.
Memory (peak)	2973 MB	Improve	<= 1000 good, > 4000 poor	budget	Peak resident memory.	Includes the in-memory recording buffer, which grows with run length (fix F15).
Telemetry overhead	0.004 % of loop step	Good	<= 1 good, > 5 poor	budget	Estimated cost of the measurement itself.	A timing wrapper per agent call; estimated by micro-benchmark, not an uninstrumented A/B run.







Left: SLAM map with the true path (blue), the estimated path (orange) and the 48 frontiers left at the end (red circles, most at or beyond the outer wall). Right: reference grid built with the true pose.

2b. Does each SLAM component help? (offline replay of this run's recording, one component removed)

The same recorded scans and odometry are replayed with one component switched off, so any difference is caused by that component. Replay check: the full replay matches the live trajectory to 0.01 cm and two identical replays are identical. 'Helps' means ATE rises by 25 % or more when it is removed; 'Not shown' means under 10 %. One recording only.

Configuration	ATE (cm)	Change vs full	Heading err (deg)	Verdict	Why
Remove: scan matching (odometry + gyro only)	264.8	+756 %	44.62	**Helps**	Without it the heading drifts to 44.6 deg and the position error grows to metres.
Remove: loop closure	54.4	+76 %	0.71	**Helps**	Closures pull accumulated drift back; without them error grows between revisits.
Remove: degeneracy filter	32.9	+6 %	0.74	**Not shown**	No measurable effect on this run (final error was slightly lower without it).
Remove: Manhattan heading prior	168.9	+446 %	18.78	**Helps**	The wall-direction prior cancels gyro heading drift; without it heading error reaches 18.8 deg.

3. Comparison with published systems

Result of the search: no published system was evaluated under our conditions, so none is directly comparable. The closest are partial matches: the A1 SLAM paper (same robot type, short paths), the ANYmal D evaluation (quadruped, long missions, real outdoors) and FARE (autonomous exploration, wheeled). Numbers below are quoted from the cited tables; 'derived' means we divided two published numbers. Differences are listed so a reader can judge.

3a. SLAM as a whole: pose and map together (partially comparable)

System	Test conditions	Pose: ATE RMSE (cm)	Map: accuracy / completion (cm) / completion ratio (%)	Tracking per frame (ms)	Mapping per frame (s)	Map memory (MB)
Ours	simulated 8-room arena, 158 m; forward RGB-D to 2D scan + gait odometry + gyro; 2D occupancy grid; CPU [this work]	31.0	12.9 / 6.4 / 54.1	10.6	0.020	0.19
MNE-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.398	1.681 / 1.509 / 97.0	22.1	0.14	24.45
Loopy-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.369	1.593 / 2.645 / 91.1	-	-	-
GO-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.404	2.091 / 2.415 / 93.2	-	-	-

ESLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.657	2.082 / 1.754 / 96.0	68.5	0.29	20.73
Co-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	1.059	2.104 / 2.082 / 93.4	54.8	0.15	26.39
Point-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.559	1.935 / 2.779 / 90.5	29.7	10.47	31.42

Reading: on both halves of SLAM we are an order of magnitude behind the neural RGB-D systems on Replica: pose error 31.0 cm against 0.4-1.1 cm, wall accuracy 12.9 cm against 1.6-2.1 cm. We are far cheaper: a map of 0.19 MB against 20-40 MB, and 0.02 s of mapping per frame against 0.14-10.5 s, on a CPU. These are not like-for-like: Replica is small single scenes with a 3D mesh map and GPU; ours is a 2D wall-cell map, 5 cm cells (which cannot resolve finer than about 2.5 cm), a forward camera with a 57 deg field of view, and gait odometry. The comparison shows where the gap is, not a ranking.

Multi-room scenes (closest scene type to ours): ATE RMSE of published systems on Replica apartments against our arena.

System	Test conditions	ATE RMSE (cm)
Ours	own 8-room arena, 158 m [this work]	31.0
MNE-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.02 / 0.59 / 0.48
ESLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.17 / 0.79 / 0.76
Loopy-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.43 / 0.60 / 0.64
ORB-SLAM3	Replica multi-room apartments 1 / 2 / 0 [1]	4.93 / 1.36 / 1.07
Swarm-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.56 / 5.61 / 1.80
CP-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.73 / 1.85 / 0.91

3b. Localization (partially comparable)

System	Sensors	Environment	Path (m)	ATE RMSE	ATE / path
Ours (this run)	forward RGB-D to 2D scan + gait odometry + gyro	simulated 10 x 10 m, 8 rooms	158	31.0 cm	0.20 %
A1 SLAM paper (Unitree A1), 4 trajectories	2D LiDAR + IMU + depth; motion-capture truth	indoor lab	5.0-5.3	A1 SLAM 4.3-14.5 cm; Cartographer 4.0-139.7 cm	0.8-2.9 % (A1 SLAM)
FAST-LIVO2, ANYmal D (GrandTour)	LiDAR + camera + IMU (Honeywell)	mostly outdoor, 7 missions	145-332	10.5 cm (mean over missions)	0.04 % (derived)
ORB-SLAM3, ANYmal D (GrandTour)	stereo (global shutter)	mostly outdoor, 7 missions	145-332	33.0 cm (mean over missions)	0.14 % (derived)
RTAB-Map, ANYmal D (GrandTour)	RGB-D (ZED2i)	mostly outdoor, 7 missions	145-332	241.3 cm (mean over successful runs only (2 of 7 failed))	1.03 % (derived)
ORB-SLAM3, ANYmal D (GrandTour)	RGB-D (ZED2i)	mostly outdoor, 7 missions	145-332	821.4 cm (mean over missions)	3.50 % (derived)

Reading: our drift per path length (0.20 %) is the same order as ORB-SLAM3 stereo (derived 0.14 %) and above FAST-LIVO2 (0.04 %); we do not claim to beat them. Not like-for-like: ours is simulation with exact ground truth, a small arena that the robot loops through repeatedly (which helps loop closure), and a 2D estimate; theirs are real, mostly outdoor missions with 3D estimates. The RGB-D configurations in that study (closest to our depth sensor) performed worst (2.4-8.2 m), so sensor type alone does not decide the result.

3c. Runtime and resources (reference only: different tasks and hardware)

System	Quantity	Value	Hardware / note
Ours	SLAM update, median	10.6 ms	CPU only, AMD Ryzen 7 7735HS with Radeon Graphics; 2D scan matching
ORB-SLAM2 (RGB-D)	tracking per frame	25.58 ms	Intel Core i7-4790, TUM fr3_office, feature-based
MNE-SLAM	tracking per frame	22.1 ms	authors' hardware, neural scene model
A1 SLAM / Cartographer	optimization per scan	1.65 / 0.96 ms	optimizer call only, not full pipeline
ORB-SLAM3 stereo (ANYmal D)	CPU / RAM	290 % / 1.16 GB	Corlito et al. Table V
Ours	CPU / RAM	86 % of one core / 1.0 GB mean (3.0 GB peak)	includes the recording buffer; simulator excluded

Reading: our SLAM step is cheaper than the camera-based systems, which is expected for 2D scan matching against feature tracking and says nothing about accuracy. Our whole control loop is not real time (section 2).

3d. Exploration speed and time (partially comparable: task similar, robot and environment differ)

System	Distance (m)	Time (s)	Average speed (m/s)	Robot / environment
Ours	158	1800 (not finished)	0.088	quadruped, forward RGB-D, 10 x 10 m arena
FARE paper: DSVP	1511	931	1.62 (derived)	wheeled, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s
FARE paper: TARE	1209	658	1.84 (derived)	wheeled, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s
FARE paper: ARiADNE	1053	610	1.73 (derived)	wheeled, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s
FARE paper: HEADER	1030	576	1.79 (derived)	wheeled, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s
FARE paper: FARE (LLM + RL)	1048	590	1.78 (derived)	wheeled, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s
ANYmal (product page)	-	-	0.75 normal walking	inspection robot; routes, not open exploration
Spot (product page)	-	-	1.6 max	needs a recorded map; Directed Exploration only reroutes
Go2 (product page)	-	-	5 max (lab)	follows a user-specified path

Reading: distances and times cannot be compared (their environments are far larger and the robot differs), and we did not finish. Speed is a property of the robot: ours (0.088 m/s) is slower than every system listed. Spot and Go2 are documented as needing a recorded map or a user-given path; the ANYmal page does not say whether it explores unknown space. None of the three product pages gives exploration time or accuracy figures.

3e. Mapping on its own (not comparable)

System	Accuracy	Completion	Completion ratio	Representation
Ours	12.9 cm	6.4 cm	54.1 %	2D occupancy grid, 5 cm cells, wall cells
MNE-SLAM (Replica)	1.681 cm	1.509 cm	96.957 %	3D neural implicit map, mesh surfaces, GPU

Reading: the metrics share names but not definitions or scenes. A 5 cm grid cannot resolve better than about 2.5 cm, so a cell-to-cell comparison with a 1.7 cm mesh error is not fair. We make no better/worse claim.

3f. Experiments that would make a fair comparison possible

(1) Run Cartographer or another open 2D SLAM on this run's recorded scans and odometry: identical data and ground truth, the only head-to-head available (F25). (2) Run a standard nearest-frontier explorer in the same arena with the same robot for time to complete (F26). (3) Repeat our run at least three times with different gyro seeds to give a spread (F18). (4) Hardware validation (F24).

4. Fixes needed

Full register with evidence and acceptance checks: docs/FIXES.md. Highest priority first:

ID	Fix	Why (evidence in this run)
F1-F3	Remove phantom frontiers, keep the blacklist, freeze room labels	48 frontiers left, most unclearable; 46 % of the path wasted; 2.2 entries per room
F4	End runs on completion or no progress, not a fixed cap	run ended by the cap
F5	Raise walking speed after F1-F3	0.088 m/s
F7-F9	Reduce local jitter and wall smear; tighten ATE	RPE 7.4 %, wall accuracy 12.9 cm, ATE 31 cm
F12-F14	Profile and remove stalls	real-time factor 0.32; loop stalls to 2.2 s

5. What this baseline can and cannot show

It shows how the current system behaves in this arena in one run. It does not show repeatability (one run), real-robot performance (simulation), independent map accuracy (reference built from the same sensor), or the contribution of mapping,

frontier selection and planning (only SLAM components were ablated). Scan-match acceptance and tracking-loss counts are diagnostic. Interventions and collisions were not measurable. Class limits marked *target* are our proposals, not published standards.

Sources

MNE-SLAM: Multi-Agent Neural SLAM for Mobile Robots (Deng et al. 2025) https://doi.org/10.1109/CVPR52734.2025.00146 | ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras (Mur-Artal & Tardos) https://arxiv.org/abs/1610.06475 | A1 SLAM: Quadruped SLAM using the A1's Onboard Sensors (Chen & Dellaert) https://arxiv.org/abs/2211.14432 | Sensor Configuration Matters: A Systematic Evaluation of Multimodal SLAM on Quadruped Robots (Corlito et al. 2026) https://arxiv.org/abs/2606.19067 | Comparison of Various SLAM Systems for Mobile Robot in an Indoor Environment (Filipenko & Afanasyev) https://arxiv.org/abs/2501.09490 | FARE: Fast-Slow Agentic Robotic Exploration (Liao et al. 2026) https://arxiv.org/abs/2601.14681 | Boston Dynamics, Spot product specifications and Spot SDK 5.2.0 autonomy documentation (GraphNav, Directed Exploration). (Boston Dynamics) https://bostondynamics.com/products/spot/ | ANYbotics, ANYmal product page and technical specifications. (ANYbotics) https://www.anybotics.com/robotics/anymal/ | Unitree Robotics, Go2 product page and specification table. (Unitree) https://www.unitree.com/go2